

KAPIL IYER

✉ k22iyer@uwaterloo.ca | 📞 +1 (647) 248-2756 | 🔗 [LinkedIn](#) | 🌐 [kapil-iyer-portfolio.vercel.app](#) | 📄 [github.com/Kapil-Iyer](#)

EDUCATION

University of Waterloo

Expected 2030

Bachelor of Mathematics Co-op (Honours Applied Math - Scientific Machine Learning and Statistics), Computing Minor

Relevant courses: Multivariable Calculus, Linear Algebra, Data Structures, Probability, Statistics, Algorithm Design

SKILLS & CERTIFICATIONS

Languages: Python, SQL, JavaScript, TypeScript, R, C

Technologies: PostgreSQL, SQLite, Supabase, React, Docker, Git, Render, Vercel, Tableau, Power BI, Excel

Libraries: PyTorch, scikit-learn, Pandas, NumPy, SciPy, ChromaDB, FastAPI, Plotly, Streamlit

Certifications: Bloomberg Market Concepts (BMC) 🐉, Bloomberg Spreadsheet Analysis (BQL) 🐉, BFF 🐉

EXPERIENCE

Ibreez Academy Inc.

May 2026 - August 2026

AI Product Engineer Intern (Co-op)

Mississauga, Canada

- Built **SIRS** 🐉, a **Python + Supabase** student-operations platform for **3,000+** students across **27** course types.
- Engineered **7+** **operational workflows** across **3** synchronized **Streamlit** apps spanning registration, payments, kiosk/live attendance, exam tracking, **ReportLab** PDFs, and real-time analytics for lead generation and revenue.

Wat Street - Waterloo Quantitative Finance Design Team

January 2026 - August 2026

*ML Engineer - **Volatility Contagion Explorer** 🐉*

Waterloo, Canada

- Collaborated on a **GAT + LSTM** system forecasting next-session volatility for **155 of 500 S&P 500 equities**.
- Implemented the **GATLayer** (**10 features** → **32D embeddings**, message passing, graph-to-LSTM interface) and built the interactive product tour for exploring forecasts, network influence, risk rankings, and shock propagation.

Dazia Consulting Inc.

November 2025 - January 2026

*AI Engineer - **FinanceBuddyAI** 🐉*

Kitchener, Canada

- Expanded a **RAG pipeline** by adding semantic chunking, indexing **1500+ pages** of CSC/IFIC content across **1,600 embeddings** per dataset in **ChromaDB**, and deploying the vector store with **Docker** for persistent storage.
- Built **8 Express.js REST APIs** with **CORS** and **rate limiting**, scaling backend across **4 AI tutoring services**.
- Implemented a **20-interaction** per-service **freemium** cap aligned with cost modeling and reduced **Gemini 2.5 Flash API** inference costs by **30%** through retrieval optimization, integrating the backend with a **React** frontend.

TECHNICAL PROJECTS

Wanderers.space (Founding Engineer)

Website 🐉

TypeScript, Next.js, React, Supabase, Gemini 2.5 Flash, Google Maps API, Vercel

- **Co-founded** Wanderers, a live campus social platform enabling University of Waterloo students to form real-life connections through spontaneous activity **bubbles**, selected and pitched at **Velocity Pitch Competition 2026**.
- Engineered the backend integration layer wiring **Next.js API routes** to **Supabase** for live feed updates and a **Gemini 2.5 Flash** pipeline for converting natural language into structured bubble creation in **under 2 seconds**.

PlainCents - Personal Finance Analytics Dashboard 🐉

Demo 🐉 | Website 🐉

Python, TypeScript, React, FastAPI, scikit-learn, SQLite, Pandas, yfinance, Power BI, Render

- Built a **full-stack personal finance** platform for multi-bank analytics, forecasting, portfolio tracking, and exports.
- Improved ML transaction categorization from **0.17 to 0.59 macro-F1** using **TF-IDF + Logistic Regression** and **human-in-the-loop** safeguards, while validating a **3-month rolling-mean** forecast through temporal backtesting.

RiskFecta - Quantitative Portfolio Intelligence Platform 🐉

Website 🐉

Python, TypeScript, React, PyTorch, scikit-learn, SciPy, Supabase, FastAPI, Plotly, Bloomberg Terminal, Vercel

- Forecasting **21-trading-day** returns across **50** equities using **Bloomberg** for OHLCV, SPX, VIX, and rates data.
- Developing parallel **LSTM + XGBoost** models with walk-forward validation and **equal-weighted ensembling** to drive **Modern Portfolio Theory**, constrained mean-variance optimization, and **Efficient Frontier** analysis.